IDC® - Bearings
by INTERPRECISE Germany

Bearings for tablet presses

INTERPRECISE Donath GmbH
Ostring 2
90587 Obermichelbach
Germany

Telefon +49-911-76630-0
Fax +49-911-76630-30

info@interprecise.de
www.idc-bearings.com



IDC® - Bearings
by INTERPRECISE Germany

Interprecise Donath GmbH with it's brand IDC® stands for flexibility and innovation, customer oriented thinking and quality focused work.
It is our philosophy to realize economical solutions for demanding customer-requests through technological know how and state of the art production equipment.

The company was founded in 1980 and is situated in the technological cluster of Herzogenaurach, well known for it's tradition and competence in bearing manufacturing.





Interprecise Donath GmbH offers a vast assortment of standard bearings based on DIN-ISO, as well as multiple variations thereof and customer specific special bearings.

Upon customer request we manufacture bearings in higher accuracy classes, with specific internal clearance, with greases selected for the application, individual materials for bearing seals, hybrid bearings as well as various surface finishes and coatings.

Our individual solutions are a perfect fit for the demanding applications of the tablet press industry.
To cater for the specific requirements of the medical Industry, our bearings come with FDA-grade seals, various coatings against corrosion and dedicated lubrication.

These modifications make it possible to use our bearings in harsh conditions, provide optimum machine uptime and offer solutions for the challenges of hygiene and product life.
Our most popular product ranges for this Industry are full complement cylindrical roller bearings, crossed roller bearings and special bearings on request.

IDC® - Bearings
by INTERPRECISE Germany

Full complement cylindrical roller bearings

Full complement cylindrical roller bearings have solid inner and outer rings. By incorporating the largest possible number of rollers, they are suitable for very heavy radial loads and are space-saving at the same time.

Moreover these bearings have a high rigidity. As a consequence of higher friction between the rollers, due to their kinematic conditions, the maximum rotational speed of full complement cylindrical roller bearings is significantly lower than the acceptable speed for caged cylindrical roller bearings.

Type NNF

This bearing family comes with sealing on both sides. The outer ring has a fix middle shoulder while the inner ring has three fix shoulders.

The two inner rings are hold together with clamping rings which can not bear thrust loads. Thus the rings have to be supported adequately to be able to carry thrust loads.

This bearing family is mainly used for winches and running wheels. The outer ring has 1mm less width compared to the inner ring. Seals made from polyurethane on both sides are seated in a groove on the inner ring and have a low pressure contact to the outer ring raceway.

In the tablet press industry these bearings are often used for the press roller.
Depending on running conditions and requirements for the machine cleaning, these bearings can be in standard version, come with a zinc-iron anti-corrosion coating, chemical nickel plated or if need be made from stainless steel.

We can also offer to combine bearing and press roller into one component, like a large size cam-follower.



IDC® - Bearings
by INTERPRECISE Germany

Full complement cylindrical roller bearings

Further families in the offering of IDC®

Type NNCL

The inner ring of this double row type has three fixed flanges, while the outer ring is flangeless. A retaining ring is inserted in the outer ring between the roller rows and keeps all bearing components together. Axial displacement of the shaft relative to the housing in both directions can be accommodated within the bearing. The retaining ring is not suitable for carrying axial loads



Type NCF

The inner ring of this bearing type has a single roller path with two fix shoulders, the outer ring has a single shoulder.

On the side without shoulder, the rollers are secured with a snapring.



Type NJG

In this single row version only the heavy design of dimension group 23 is produced.

These bearings are mainly used for heavy duty slow moving applications. These bearings have an outer ring with two shoulders and an inner ring with a single shoulder.

Contrary to all other full complement cylindrical roller bearings, this family has a self-retaining roller complement.
When removing the inner ring, rollers will still stay in place and not fall out, largely simplifying the mounting procedure.



IDC® - Bearings
by INTERPRECISE Germany

Type NNCF

This type is a double row cylindrical roller bearing. The inner ring has three fixed flanges.
The outer ring has only one fixed flange. On the flangeless side of the outer ring there is a retaining ring inserted which keeps all bearing components together. These bearings are able to compensate for length changes of shafts as a consequence of thermal expansion within the range of the axial displacement ability of their outer ring.



Type NNC

These double row cylindrical roller bearings have an inner ring with three shoulders and an outer ring with two shoulders on the outside.

The outer ring is split and halves retained together with fixing elements.





Further information about our range of full complement cylindrical roller bearings can be found in our brochures available in the download-section of our homepage www.interprecise.de.

IDC® - Bearings
by INTERPRECISE Germany

Crossed roller bearings

Because of their technical properties, IDC Crossed roller bearings are ideally suited for precision applications. They unite high running accuracy and stiffness in a small envelope.

Due to their specific construction with two sets of rollers and raceways brought together at right angles, these bearings can carry radial loads, axial loads in both directions, high overturning moments and any combination of those in a single bearing. As a consequence the construction can often be reduced to use a single bearing.

Series SX

Bearings of our SX-range consist of a one-piece inner ring, cylindrical rollers, spacers and a split outer ring, held together with fixing rings.



IDC® - Bearings
by INTERPRECISE Germany

Series XU / XSU

These bearings have a massive inner and outer ring, cylindrical rollers, spacers and a filling plug with locking pin.

Usually they are manufactured for high rigidity, with preload or with zero clearance. They are usually fitted with a contact seal or with labyrinth seals.

Outer and inner ring can directly be bolted to the surrounding structure. For a better fitting they can be equipped with a centring diameter.

The attachment bores can be made through-bored, with thread or blind hole.



Depending of application needs, the seals can be made in NBR, FKM or silicone.

We can use food-grade grease NSF H1, as well as coat the rings with chemical nickel.

IDC® - Bearings
by INTERPRECISE Germany

Crossed roller bearings

Further types of crossed roller bearings in our range

Series XV

Bearings of XV design have a massive outer ring, a split inner ring, cylindrical rollers, separators and labyrinth seals.

The inner ring is centred on the shaft and axially positioned and secured against a shoulder with a precision locknut.

Thus the bearing clearance can be adjusted between play and preload. The outer ring is centred in, and bolted to the housing.



Series XB

Bearings of this family have a solid inner ring, a split outer ring cylindrical rollers and spacers.

The bearing clearance is manufactured to the customers specification (ranges between clearance and preload). If needed the bearings will be fitted with rubber seals.

The bearings are fixed to the surrounding parts, similarly as the SX-type bearings, with clamping rings.



Further information about our range of crossed roller bearings are available in our brochure in the download-area of our homepage www.interprecise.de.



IDC® - Bearings
by INTERPRECISE Germany

Research and development traditionally play an important role at Interprecise.

During the design process all critical parameters of the bearing, concerning function, customer value and reliability of the product are defined, thus determining all further costs for the product life-cycle.

A thorough communication with our customers ensures that product requirements, application conditions and customer objectives are well understood and precisely described. Our highly qualified engineers and technicians analyse the information.

Concepts are drafted with consideration for maximum customer benefit and economical aspects while meeting the product requirements.



Drafts and designs are made by our skilled employees with a 3D-design program. Results of the design-process are verified and evaluated through methods of finite elements, multi- body-simulation and analytical calculation methods. They are validated through prototyping and test-samples.

Changes during the product development can be traced through our product-management-system at any time.

IDC® - Bearings
by INTERPRECISE Germany

To achieve a high level of quality, flexibility and efficiency, we execute the main operations turning and grinding internally.

To achieve our objectives we work with modern state of the art CNC equipment.



All of our activities in the company are subject to our commitment to quality.

Quality assurance is conducted in each stage and step of the product development.

Our employees use the latest measuring equipment available in the industry.

All process' of the company are certified to ISO 9001:2015.



IDC® - Bearings
by INTERPRECISE Germany

INTERPRECISE Donath GmbH
Ostring 2
90587 Obermichelbach
Germany

Phone +49-911-76630-0
Fax +49-911-76630-30

info@interprecise.de
www.idc-bearings.com